

Alexander X Yu

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Education

University of California, Los Angeles

B.S. in Computer Science

Los Angeles, CA

September 2019 - June 2023

The Harker School

High School

GPA: 4.0 (unweighted), 4.54 (weighted)

Relevant coursework: APCS, Data Structures, Expert Systems, Computer Architecture, Multivariate Calculus, Differential Equations

San Jose, CA

August 2015 - May 2019

Experience

Li Lab

UCSF Department of Biochemistry
and Biophysics

Research Intern

June 2018 - Sep 2018, June 2019 - Present

- Used yeast as a model system to examine causal links between the cell cycle regulator WHI5 and aging process
- Used machine learning algorithms and image/data analysis techniques (Tensorflow, Scikit, SciPy, Pandas, Matplotlib, etc.) to investigate biomolecular mechanisms of aging
- Built computational pipelines to analyze physical differences (body mass, metabolic rate) and genomic differences (Gene Ontology, positive selection in genes, etc.) between mammalian species that may lead to variation in lifespan

SF-Marin Food Bank

Volunteer

San Francisco, CA

April 2016 - May 2019

- Sorted fruits and packaged meals for the elderly

Skills

Programming Languages: Java, Python, C++, Bash, HTML, CSS, Jess/LISP, R, SQL, Javascript, and C

Leadership Positions: Harker Swim Team Captain (2019), Harker Orchestra Assistant Principal Cellist (2018-19)

Projects

SMEP

JESS, Java

An expert system that helps to choose events to enter for a given swim meet.

Music Scroll

Python

A sheet music viewer that automatically switches pages hands-free.

GO Analysis

Bash, Python, R

A package that analyzes Gene Ontology terms and positively selected genes in mammalian species.

Awards

2019 **National Merit Scholarship Winner**, Recognized for top PSAT score nationwide.

2019 **Presidential Scholar Nominee**, Recognized for top SAT score and academic standing nationwide.

2016-2017 **USA Computing Olympiad Silver Level**, Awarded to those who complete bronze level problems.